```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded=files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Superstore.csv to Superstore.csv
```

```
df = pd.read_csv('Superstore.csv', encoding='latin1')
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region
0	1	CA-2013-152156	09-11-2013	12-11-2013	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South
1	2	CA-2013-152156	09-11-2013	12-11-2013	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South
2	3	CA-2013-138688	13-06-2013	17-06-2013	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West
3	4	US-2012-108966	11-10-2012	18-10-2012	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South
4	5	US-2012-108966	11-10-2012	18-10-2012	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South

5 rows × 21 columns

```
#DATA CLEANING
print("\n Dataset info:")
print(df.info())
```

```
 Dataset info:
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Row ID         9994 non-null   int64  
 1   Order ID       9994 non-null   object 
 2   Order Date     9994 non-null   object 
 3   Ship Date      9994 non-null   object 
 4   Ship Mode      9994 non-null   object 
 5   Customer ID    9994 non-null   object 
 6   Customer Name  9994 non-null   object 
 7   Segment        9994 non-null   object 
 8   Country        9994 non-null   object 
 9   City           9994 non-null   object 
 10  State          9994 non-null   object 
 11  Postal Code    9994 non-null   int64  
 12  Region         9994 non-null   object 
 13  Product ID     9994 non-null   object 
 14  Category       9994 non-null   object 
 15  Sub-Category   9994 non-null   object 
 16  Product Name   9994 non-null   object 
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64  
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
None
```

```
print(df.describe())
print("\n missing values:")
print(df.isnull().sum())
```

```
            Row ID   Postal Code        Sales     Quantity     Discount  \
count  9994.000000   9994.000000  9994.000000  9994.000000  9994.000000   
mean   4997.500000  55190.379428   229.858001     3.789574     0.156203   
std    2885.163629  32063.693350   623.245101     2.225110     0.206452   
min       1.000000   1040.000000     0.444000     1.000000     0.000000   
25%    2499.250000  23223.000000    17.280000     2.000000     0.000000   
50%    4997.500000  56430.500000    54.490000     3.000000     0.200000   
```

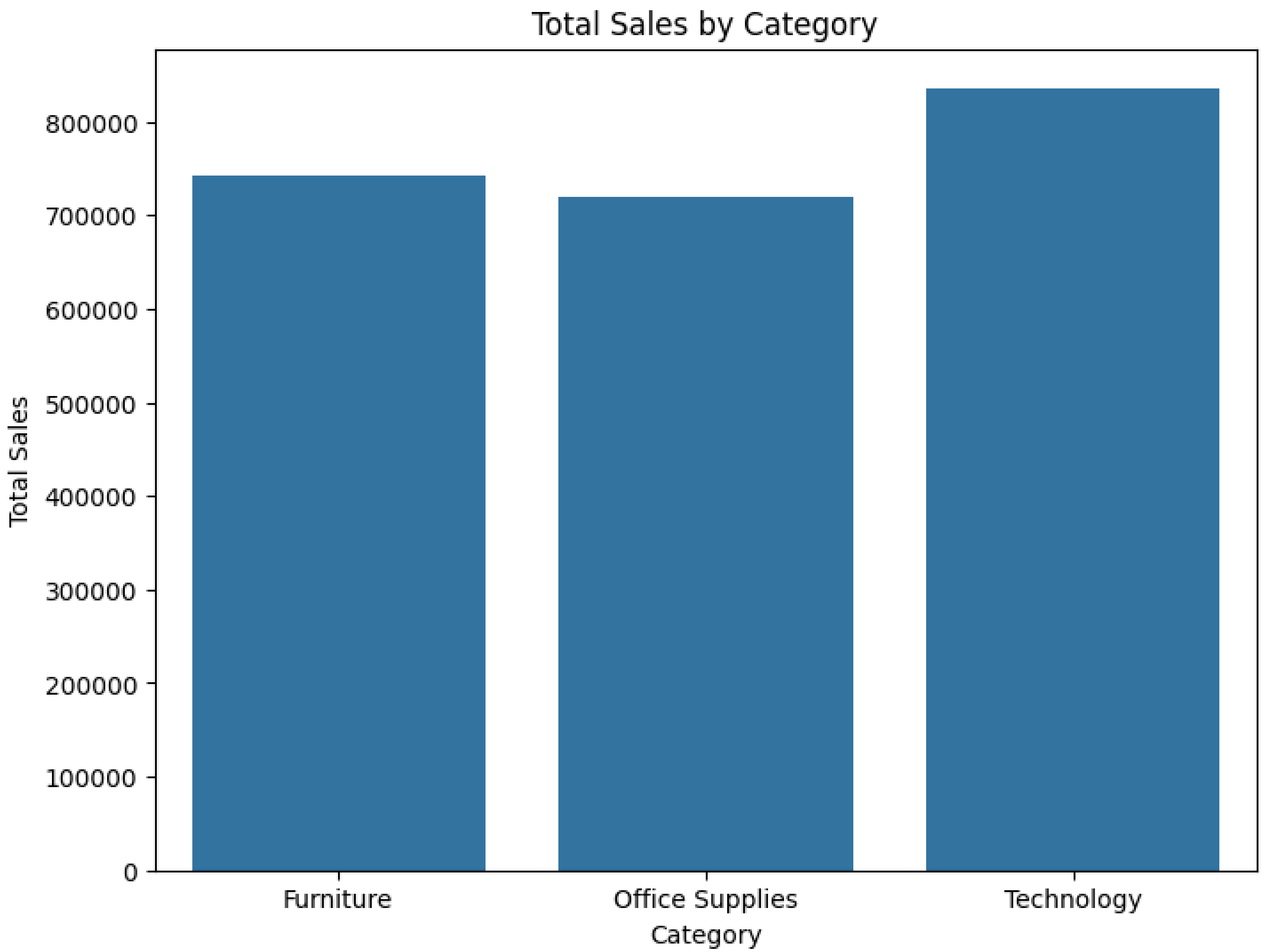
```
75%    7495.750000  90008.000000    209.940000     5.000000     0.200000
max    9994.000000  99301.000000  22638.480000    14.000000     0.800000

            Profit
count  9994.000000
mean     28.656896
std     234.260108
min   -6599.978000
25%       1.728750
50%       8.666500
75%      29.364000
max    8399.976000

 missing values:
Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code      0
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
Quantity         0
Discount         0
Profit           0
dtype: int64
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
df['Ship Date'] = pd.to_datetime(df['Ship Date'], dayfirst=True)
```

```
df['Shipping Days'] = (df['Ship Date'] - df['Order Date']).dt.days
print(df[['Order Date','Ship Date',]])
```

```
     Order Date  Ship Date
0    2013-11-09 2013-11-12
1    2013-11-09 2013-11-12
2    2013-06-13 2013-06-17
3    2012-10-11 2012-10-18
4    2012-10-11 2012-10-18
...         ...        ...
9989 2011-01-22 2011-01-24
9990 2014-02-27 2014-03-04
9991 2014-02-27 2014-03-04
9992 2014-02-27 2014-03-04
9993 2014-05-05 2014-05-10

[9994 rows x 2 columns]
```

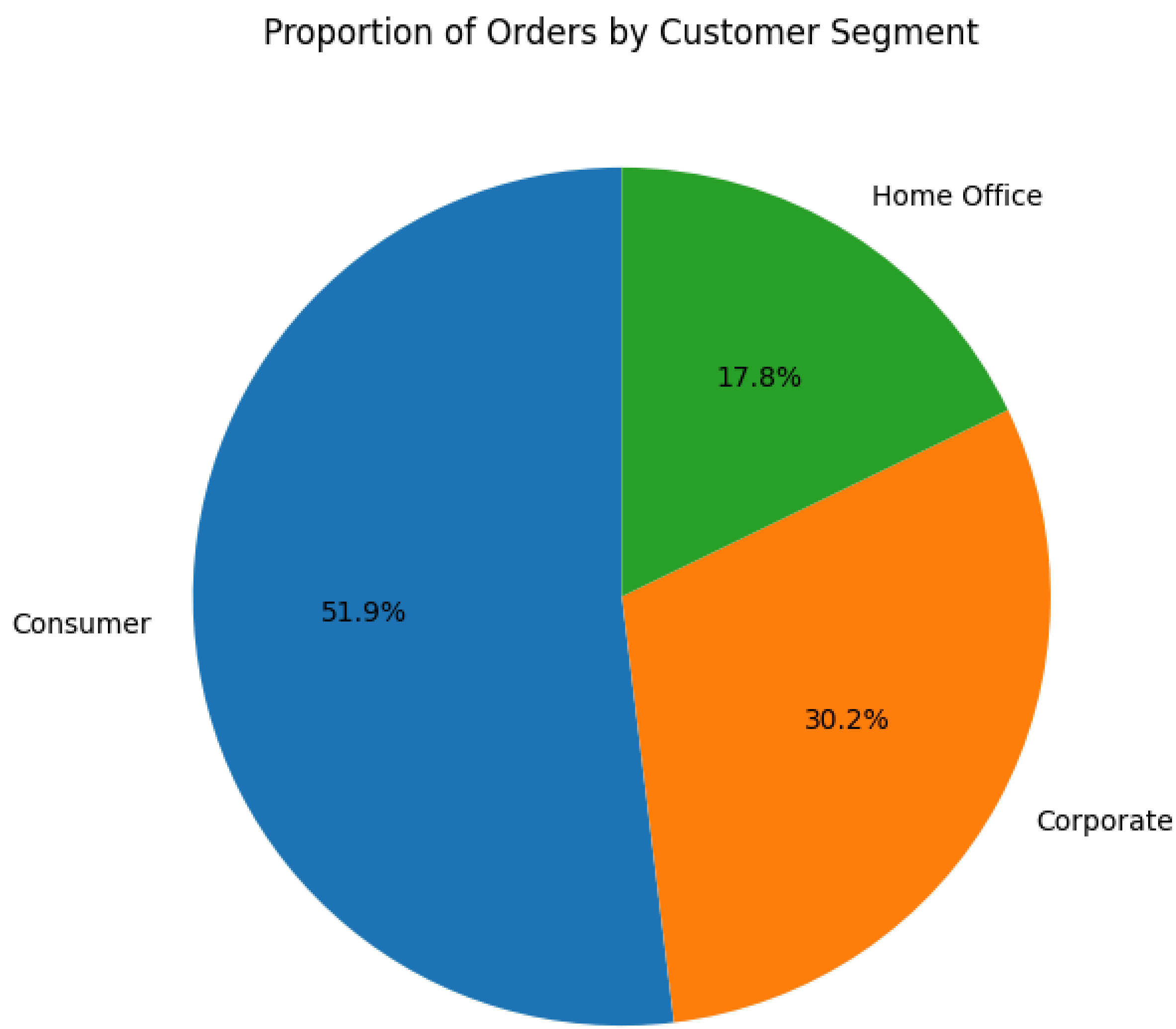
```
# DATA VISUALIZATION

# Graph 1: Sales by Category
plt.figure(figsize=(8,6))
sns.barplot(data=df, x='Category', y='Sales', estimator=sum, errorbar=None)
plt.title('Total Sales by Category')
plt.ylabel('Total Sales')
plt.xticks(rotation=0)
plt.show()
print("Observation: Technology category usually leads in total sales, followed by Furniture and Offic
```




```
Observation: Technology category usually leads in total sales, followed by Furniture and Office Suppli
```

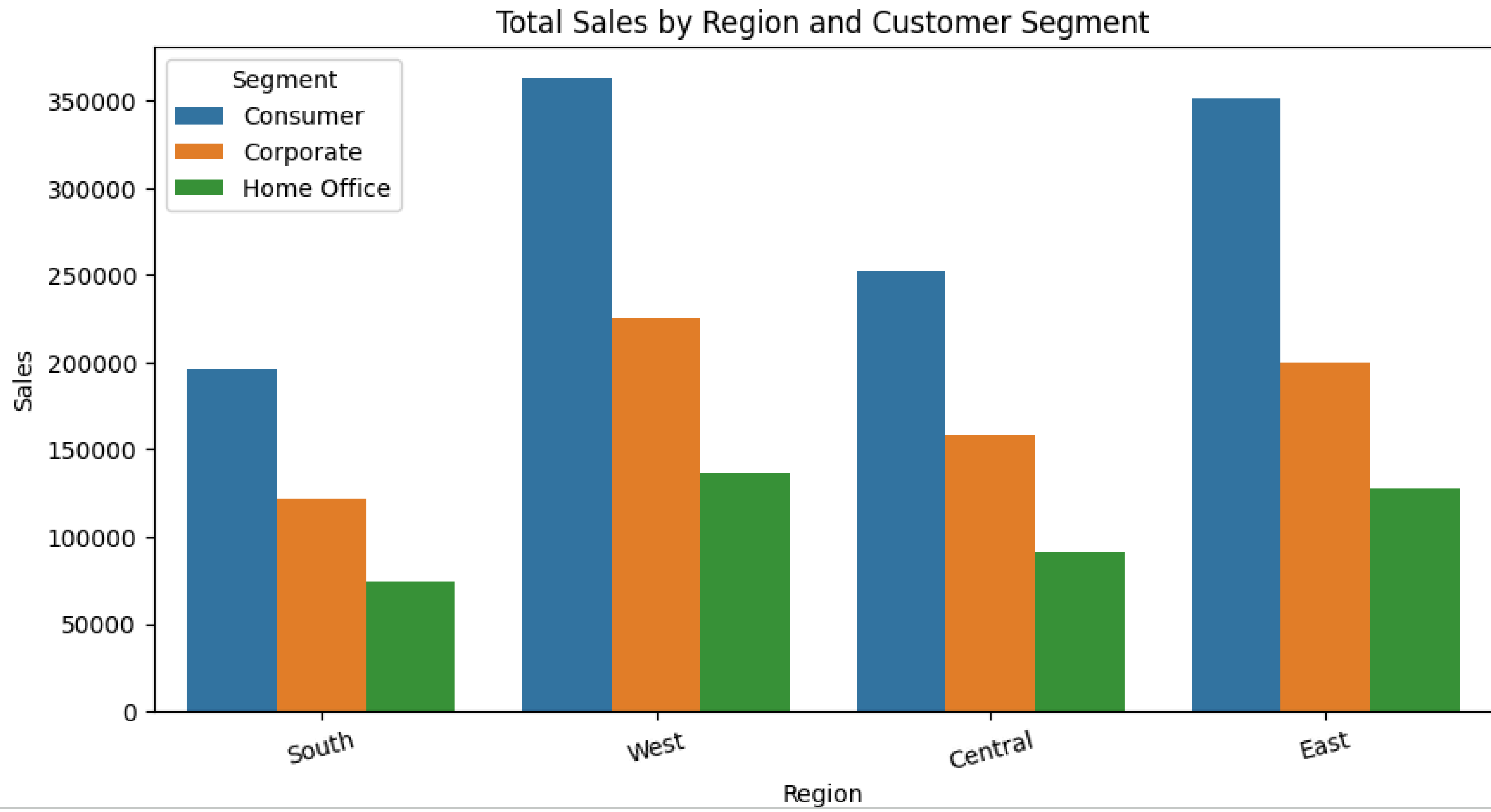
```
plt.figure(figsize=(7,7))
segment_counts = df['Segment'].value_counts()
plt. pie(segment_counts, labels=segment_counts.index, autopct='%1.1f%%', startangle=90)
plt.title('Proportion of Orders by Customer Segment')
```

```
plt.ylabel('')
plt.show()
```



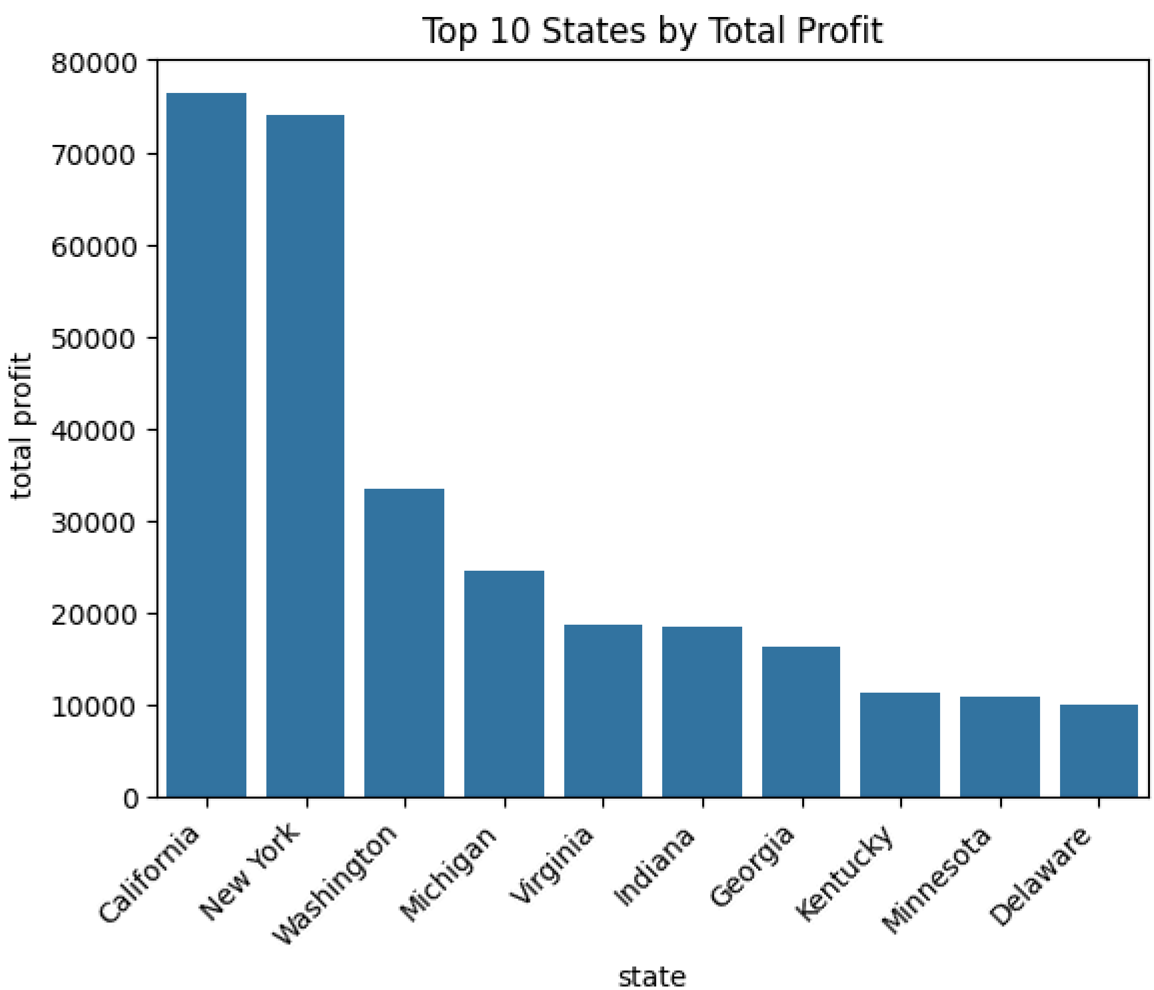

```
# Graph 3: Sales by Region and Segment
plt.figure(figsize=(10,5))
```

```
sns.barplot(data=df, x='Region', y='Sales', hue='Segment', estimator=sum, errorbar=None)
plt.title('Total Sales by Region and Customer Segment')
plt.xticks(rotation=15)
plt.show()
print(" Observation: Consumer segment contributes highest sales across all regions")
```




```
top_states = df.groupby('State')['Profit'].sum().sort_values(ascending=False).head(10)
sns.barplot(y=top_states.values, x=top_states.index)
plt.xticks(rotation=45,ha='right')
plt.title('Top 10 States by Total Profit')
plt.xlabel('state')
```

```
plt.ylabel('total profit')
```




```
# FINAL CONCLUSION
"""Key Insights from Superstore Sales EDA:

1. Category Performance: Technology drives highest sales, but Furniture has profit issues due
2. Discount Impact: Orders with discount > 0.2 are highly likely to result in loss. Discount
3. Regional Trends: West region leads in sales and profit. Central region underperforms.
4. Customer Segments: Consumer segment is the largest contributor to sales across all regions
5. Loss-Making Products: Sub-categories like Tables and Bookcases have negative total profit
```